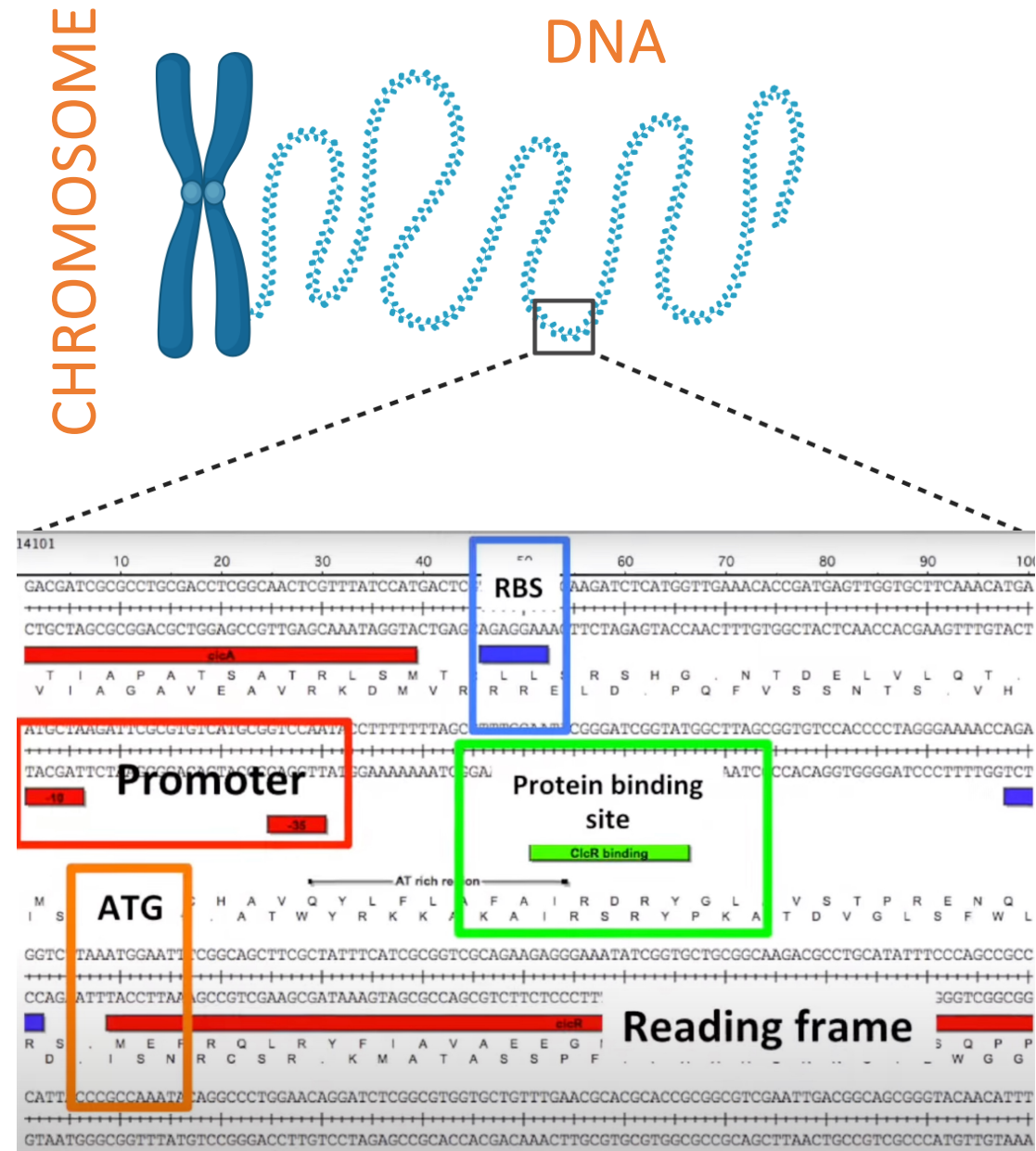
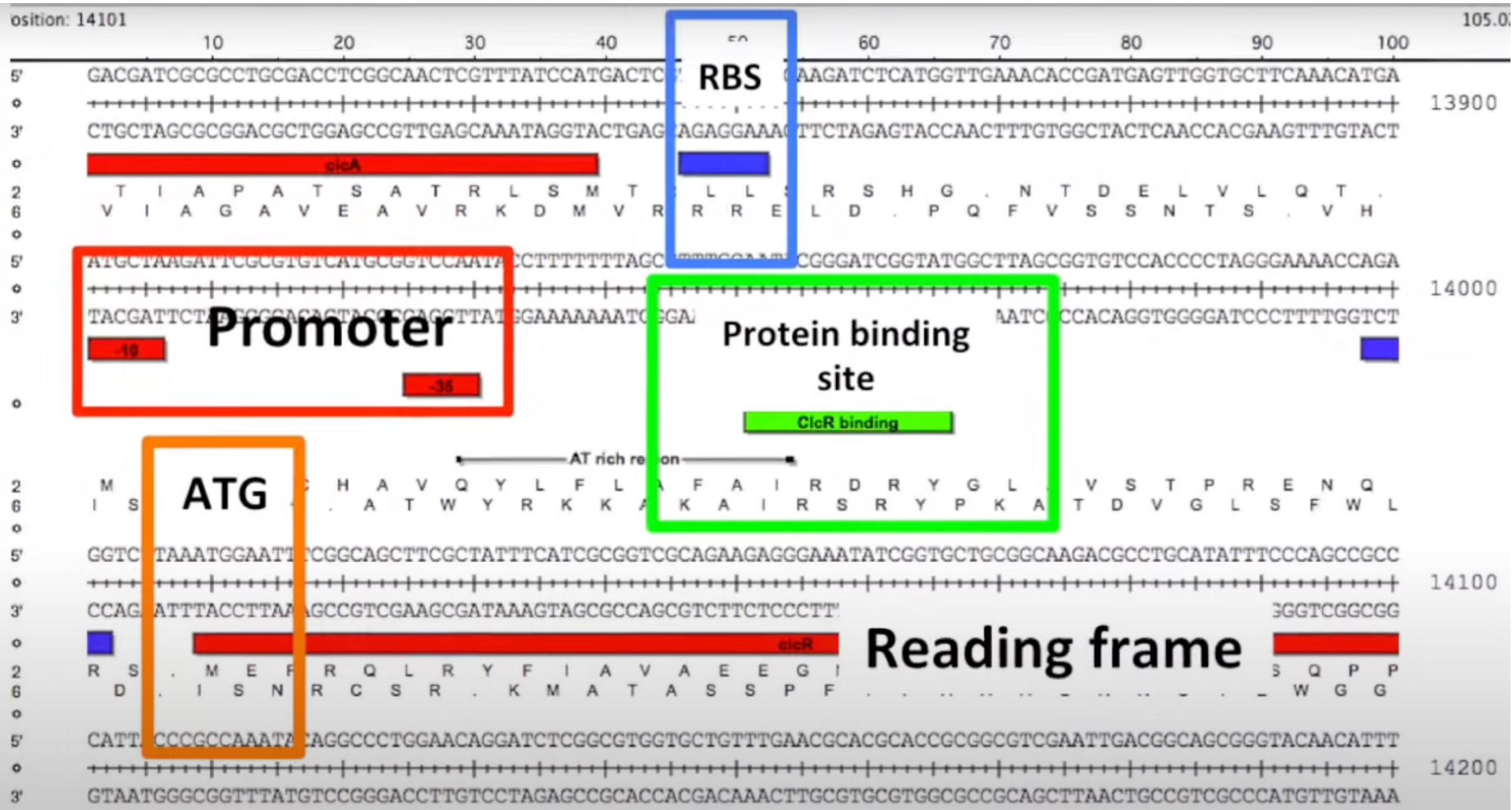


Genetic circuits and logic gates in the regulation of gene expression



Sequence analysis

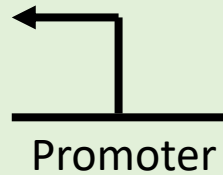


Different parts of DNA sequence have different functions

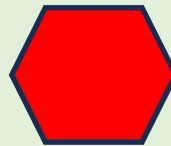
Can act as Ribosome binding sites, promoters, protein-coding frames, Protein binding sites

Circuit parts

DNA construction bricks



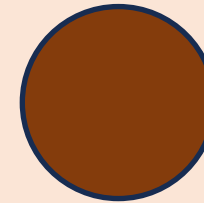
Regulatory
region



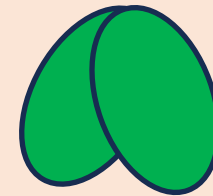
Termination
sequence

Protein parts

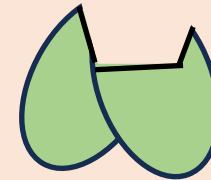
Structural
protein



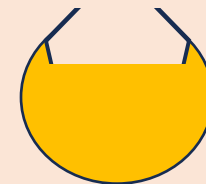
Regulator



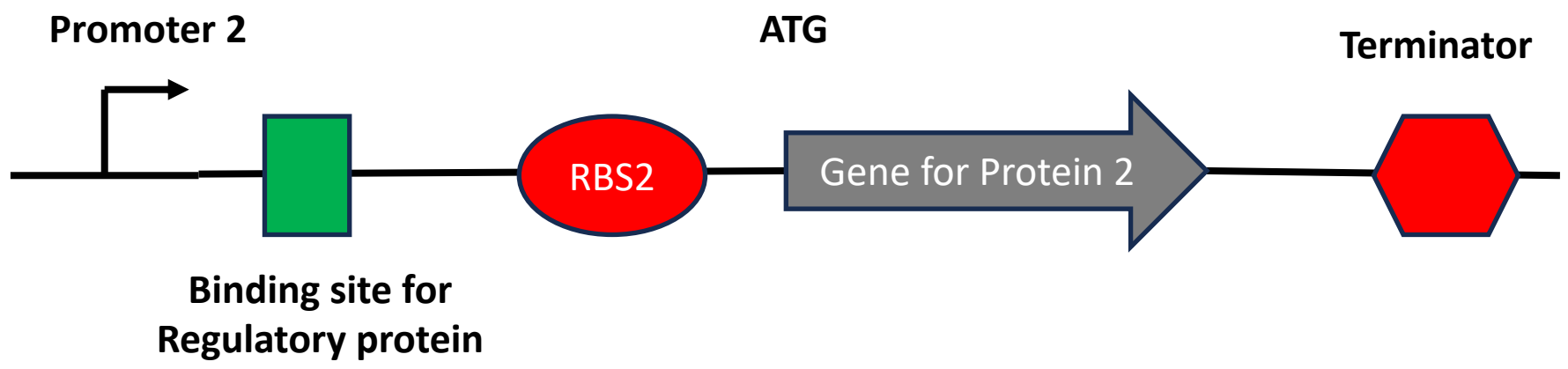
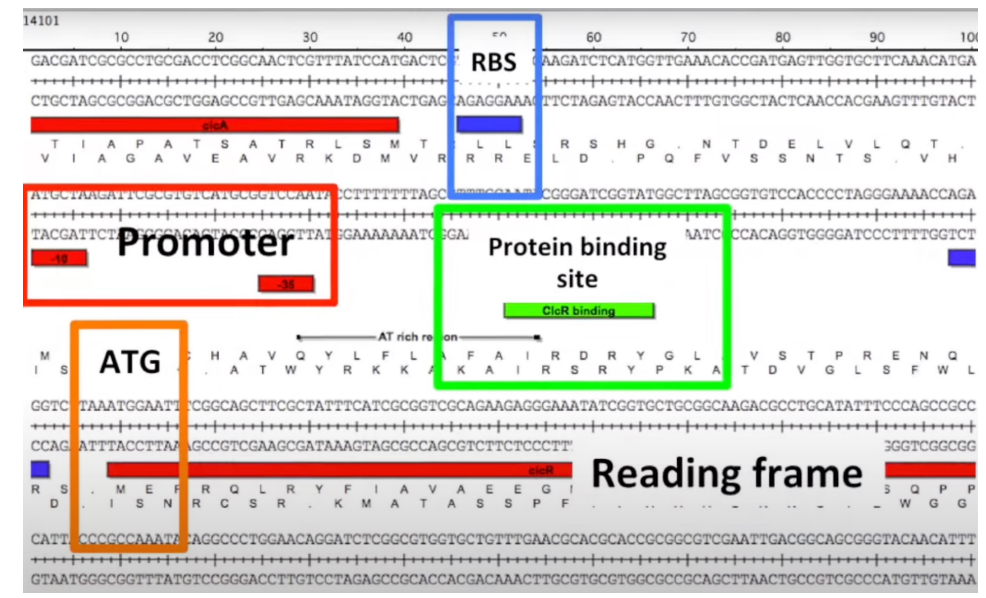
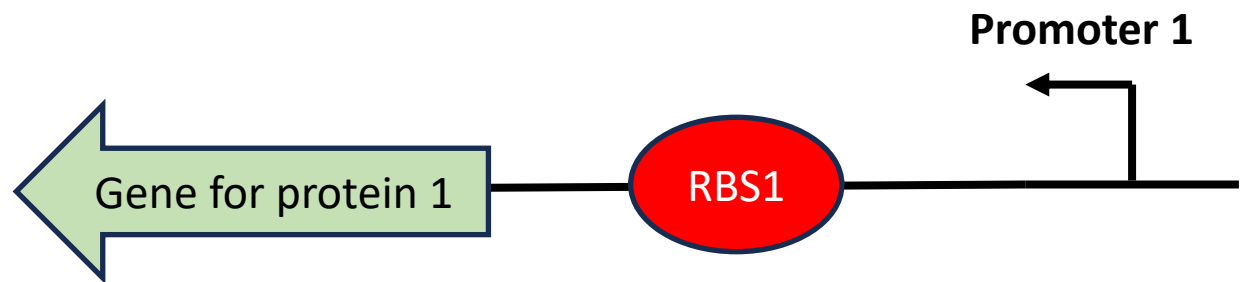
Transcription
factor



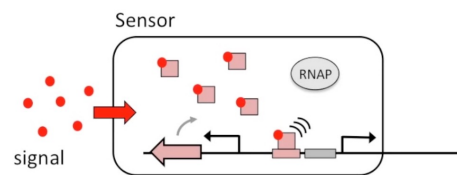
Sensory
protein



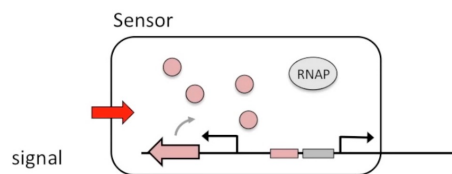
From DNA Sequence to “Circuit”



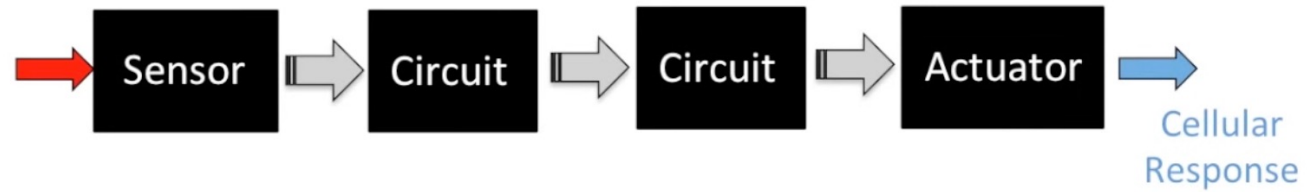
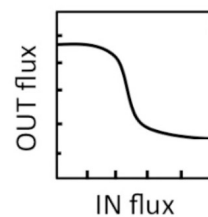
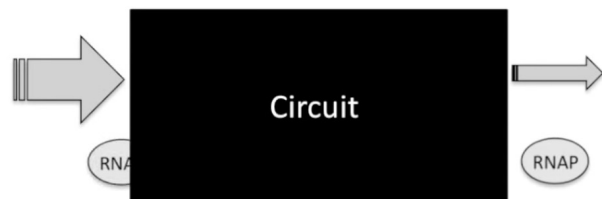
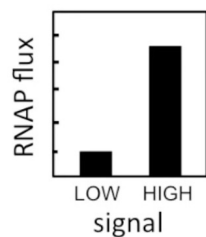
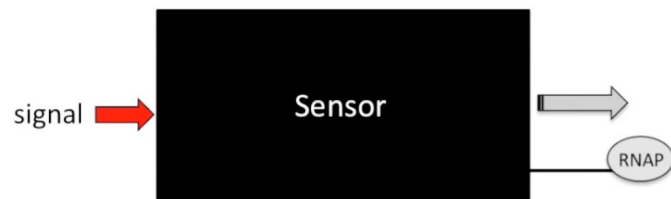
These DNA parts are arranged as a circuit



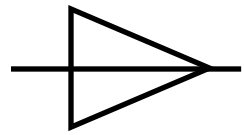
High



Low

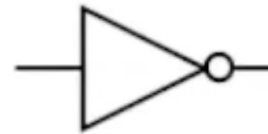


Common logic gates



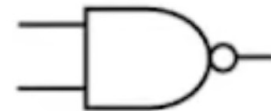
YES

Input	Output
I	F
0	0
1	1



NOT

Input	Output
I	F
0	1
1	0



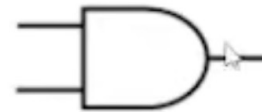
NAND

Inputs		Output
A	B	F
0	0	1
1	0	1
0	1	1
1	1	0



NOR

Inputs		Output
A	B	F
0	0	1
1	0	0
0	1	0
1	1	0



AND

Inputs		Output
A	B	F
0	0	0
1	0	0
0	1	0
1	1	1



OR

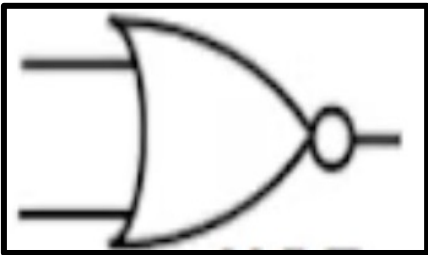
Inputs		Output
A	B	F
0	0	0
1	0	1
0	1	1
1	1	1

Apollo 11 guidance system

5600 Triple-input
NOR gates

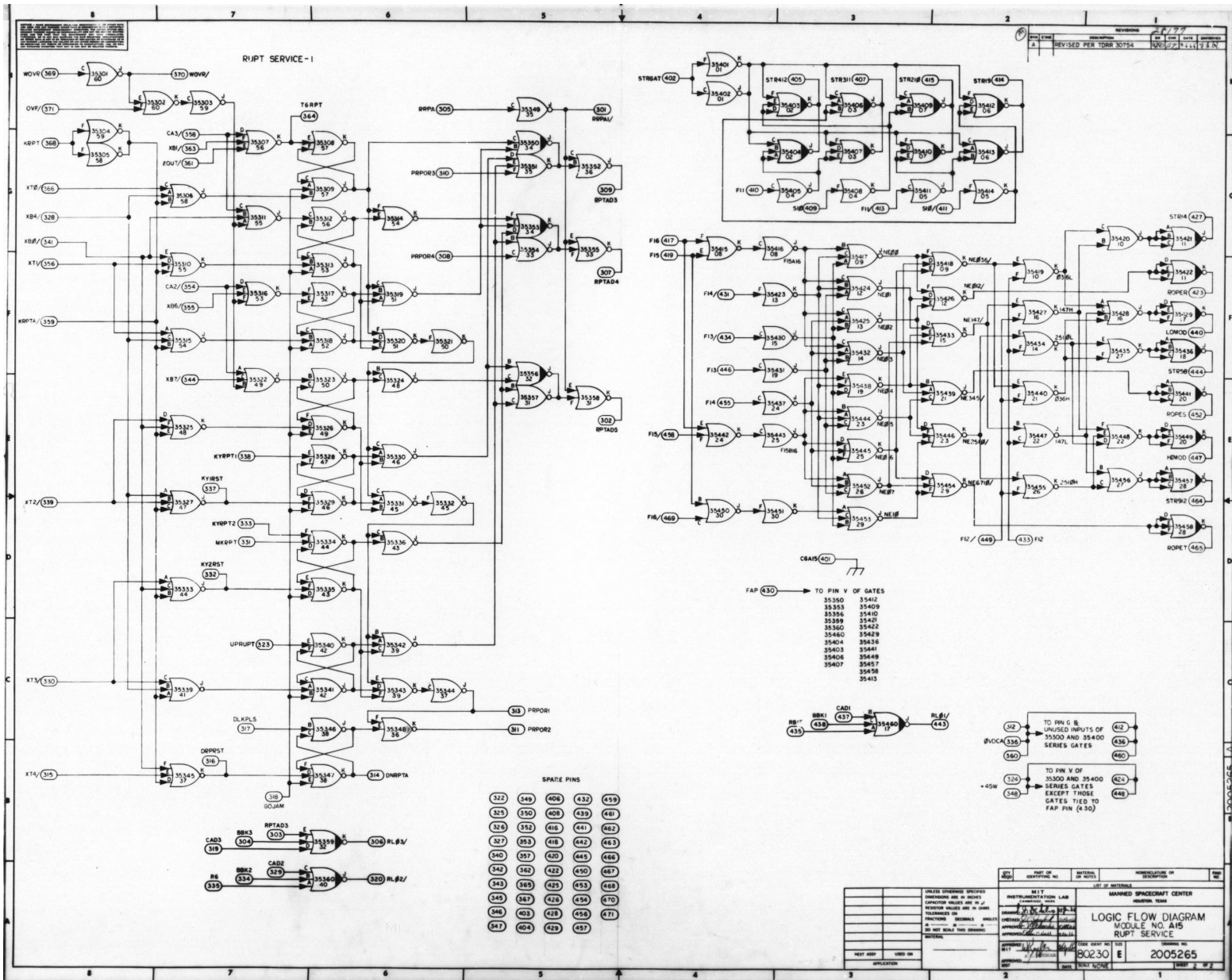
NOR Gate

IN_A
IN_B



Out

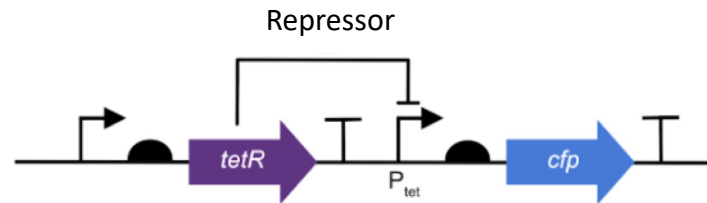
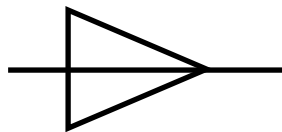
Inputs		Output
A	B	F
0	0	1
1	0	0
0	1	0
1	1	0



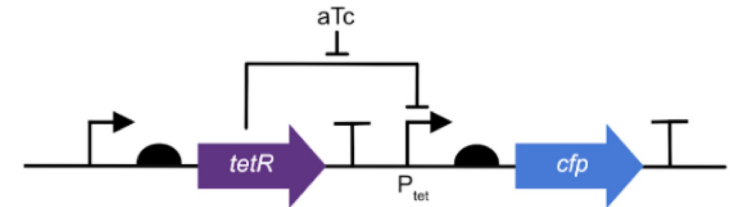
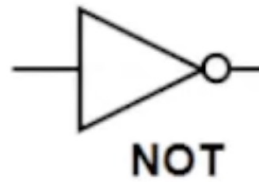
Exercise



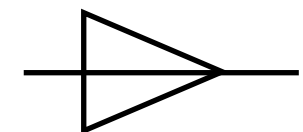
Promoter	CFP
0	
1	



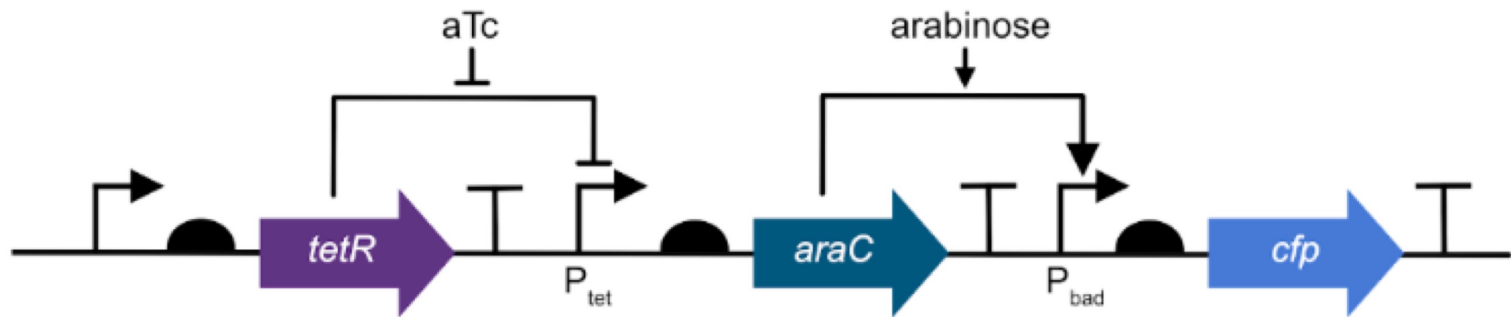
Repressor	CFP
0	
1	



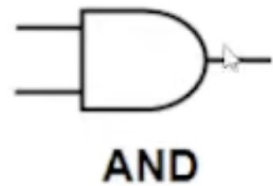
aTc	CFP
0	
1	



Exercise



aTc	arabinose	CFP
0	0	0
0	1	0
1	0	0
1	1	1



Inputs		Output
A	B	F
0	0	0
1	0	0
0	1	0
1	1	1

Exercise



Truth table

Inputs		Output
A	B	F
0	0	1
1	0	0
0	1	0
1	1	0



Logic gate

